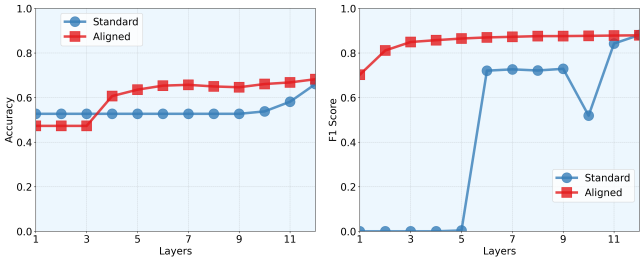


(a) RTE (b) QQP

Figure 21: Layerwise accuracy for **AlignedBERT** and BERT and on RTE and QQP datasets of the GLUE benchmark.

C.2 Aligned Training on Language Models

In this section, we provide more details about the datasets and computational resources used. The datasets (GLUE, and Wikitext-103) are publicly available under the MIT license. For NLP tasks, we used a single RTX A5000 GPUs with 24GB of memory. Then, we will introduce the experimental setup for AlignedBERT and AlignedGPT models.

Setup in AlignedBERT The General Language Understanding Evaluation (GLUE) benchmark comprises nine tasks for assessing natural language understanding. In our AlignedBERT experiments on the GLUE dataset, we used a sequence length of 256. We employed AdamW for optimization with an initial learning rate of 2e-5, and a batch size of 32. Each task underwent fine-tuning for three epochs. The WikiText-103 language modeling dataset consists of over 100 million tokens extracted from Wikipedia's verified good and featured articles. For AlignedGPT experiments on the WikiText-103 dataset, we maintained the sequence length at 256 and used AdamW with an initial learning rate of 2e-5. In this case, we set the batch size to 8. More experiment results on RTE and QQP dataset can be found in Figure 21.

Setup in AlignedGPT. Our aligned training method can be used with any transformer-based language models. In this study, we evaluated our method using the GPT-2 model. We finetune the GPT2 models using aligned training methods and then use intermediate layers of GPT2 to generate the texts.

- **Model and Baselines** We finetune GPT-2 on the Wikitext-103 dataset with the proposed objective $\mathcal{L}_{\text{aligned}}$ for $40k$ training steps and generate the text continuation with nucleus sampling [63] with $p = 0.95$ decoding methods. For the standard baseline model, we finetune the model with CE loss $\mathcal{L}_{\text{MLE}}$. The model is finetuned using a single 24G RTX A5000 GPU for 70 hours.
- **Evaluations** Following [60], we evaluate the model from two perspectives: (1) language modeling quality, assessing the inherent quality of the model, and (2) generation quality, measuring the quality of the text the model produces. In assessing language modeling quality, we calculate the prediction accuracy and perplexity of each layer. When evaluating generation quality, we measure the similarity between the prompt text and generated text using coherence. We employ generation repetition to gauge the diversity of the generated text. The metrics are defined as follows,
  - **Prediction Accuracy** The accuracy is computed on the Wikitext-103 test set as,

$$\textbf{Acc} = \frac{1}{D}\sum_{i=1}^{D}\sum_{i=1}^{n} \mathbb{1}[\arg\max p_\theta(x|\boldsymbol{x}_{<i}) = x_i] \tag{20}$$

    where the D is the number of samples in the test dataset.
  - **Perplexity** The perplexity is computed on the test set of Wikitext-103. It's computed as the exponential of the test loss.
  - **Coherence** Coherence measures the relevance between the prefix text and the generated text. We apply the advanced sentence embedding method, SimCSE [64], to measure the semantic coherence or consistency between the prefix and the generated text. The coherence score is defined as follows,

$$\textbf{Coherence} = \boldsymbol{h}_{\boldsymbol{x}}^T \boldsymbol{h}_{\hat{\boldsymbol{x}}} / \|\boldsymbol{h}_{\boldsymbol{x}}\| \|\boldsymbol{h}_{\hat{\boldsymbol{x}}}\| \tag{21}$$

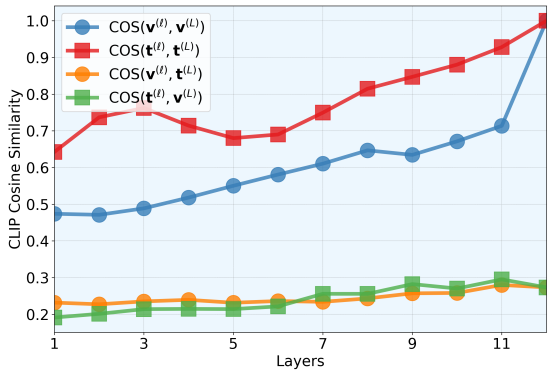


Figure 22: Illustration of cosine similarity between the hidden layer to the last layer for within and cross modalities in a 12-layer CLIP-B/32.

where $\boldsymbol{x}$ is the prefix text and $\hat{\boldsymbol{x}}$ is the generated text and $\boldsymbol{h}_{\boldsymbol{x}} = \text{SimCSE}(\boldsymbol{x})$ and $\boldsymbol{h}_{\hat{\boldsymbol{x}}} = \text{SimCSE}(\hat{\boldsymbol{x}})$. Higher coherence means more correlation to the given prompt.

– **Diversity** Diversity measures the occurrence of generation at different n-gram levels. It is defined as:

$$\textbf{Diversity} = \prod_{n=2}^{4} \frac{|\text{unique n-grams}(\hat{\boldsymbol{x}})|}{|\text{total n-grams}(\hat{\boldsymbol{x}})|} \tag{22}$$

A higher diversity score suggests fewer repeated words in the generated text.

D Additional Experiments on Multi-Modality Models

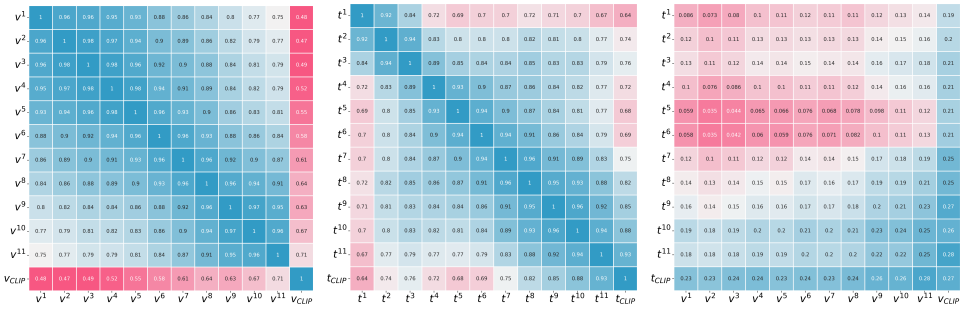
In this section, we demonstrate that the observation in Section 2.1 can also extend to multi-modality models such as the pre-trained CLIP models [65]. Given that the CLIP model comprises both a vision encoder and a text encoder, we evaluate the cosine similarity within each modality (vision or text) and across modalities (between the vision encoder and text encoder). This comprehensive analysis highlights the robustness of the observed phenomena across diverse components of the CLIP architecture.

Specifically, given a pretrained CLIP model with vision encoder f_{vision} and text encoder f_{text}, we extract the ℓ-th layer vision feature $\hat{\boldsymbol{v}}^{(\ell)} \in \mathbb{R}^{768}$ by taking the corresponding [CLS] token outputs of ℓ-th layer from f_{vision}; similarly, take the ℓ-th layer vision feature $\hat{\boldsymbol{t}}^{(\ell)} \in \mathbb{R}^{512}$ from the [EOS] token outputs of f_{text}. Since CLIP projects both the vision features and text features to the same embedding space through a vision projection matrix $\boldsymbol{W}_{\text{vision}} \in \mathbb{R}^{768\times 512}$ (followed by a layer normalization LN) and a text projection matrix $\boldsymbol{W}_{\text{text}} \in \mathbb{R}^{512\times 512}$ (also followed by a layer normalization LN), we also apply these projection matrices to the hidden layer features to obtain

$$\boldsymbol{v}^{(\ell)} = \text{LN}(\boldsymbol{W}_{\text{vision}}\hat{\boldsymbol{v}}^{(\ell)}) \in \mathbb{R}^{512}, \quad \boldsymbol{t}^{(\ell)} = \text{LN}(\boldsymbol{W}_{\text{text}}\hat{\boldsymbol{t}}^{(\ell)}) \in \mathbb{R}^{512}. \tag{23}$$

To measure within-modality cosine similarity, we calculate layer-wise similarity within the vision encoder as $\text{COS}(\boldsymbol{v}^{(\ell)}, \boldsymbol{v}^{(\ell')})$ and within the text encoder as $\text{COS}(\boldsymbol{t}^{(\ell)}, \boldsymbol{t}^{(\ell')})$. Similarly, for layer-wise cross-modality cosine similarity, we evaluate the relationships between the vision encoder and text encoder using $\text{COS}(\boldsymbol{v}^{(\ell)}, \boldsymbol{t}^{(\ell')})$ and $\text{COS}(\boldsymbol{t}^{(\ell)}, \boldsymbol{v}^{(\ell')})$.

Figure 22 plots the layer-wise within-modality similarity and cross-modality similarity by comparing the hidden-layer features to the last-layer features, while Figure 23 plots all the pair-wise results. All experiments are conducted on the CIFAR10 validation dataset, where the text input to the text encoder is: "This is a photo of a {label}". From both figures, we observe a clear pattern of progressively increasing layer-wise representational similarity in both the vision encoder and the text encoder. For cross-modality similarity, we note that the last-layer vision and text representations are not perfectly aligned (e.g., $\text{COS}(\boldsymbol{t}^{(L)}, \boldsymbol{v}^{(L)})$ is not close to 1), a phenomenon commonly referred to



(a) $\mathrm{COS}(\boldsymbol{v}^{\ell}, \boldsymbol{v}^{\ell'})$ (b) $\mathrm{COS}(\boldsymbol{t}^{\ell}, \boldsymbol{t}^{\ell'})$ (c) $\mathrm{COS}(\boldsymbol{v}^{\ell}, \boldsymbol{t}^{\ell'})$

Figure 23: Layer-wise cosine similarities within (a) vision encoder, (b) text encoder, and (c) cross vision and text encoder for a pretrained 12-layers CLIP-B/32.

as the *modality gap*, which has been consistently observed across various multi-modal models [66]. Despite this, we also observe a progressive increase in layer-wise representation similarity across modalities. Together, these results highlight a distinct trend of progressively increasing layer-wise representational similarity within and across modalities.